

ELECTRONICS LABORATORY

Experiment No:	3
Experiment Name:	Introduction to Bipolar Junction Transistors
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1. OBJECTIVES

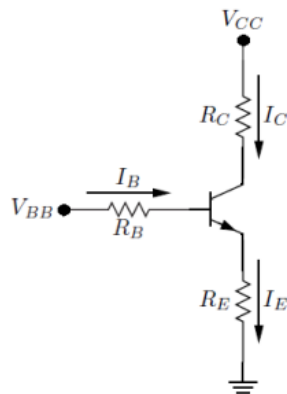
In this experiment, our aim is to examine the basic functions of Bjt..

2. EXPERIMENTAL RESULTS

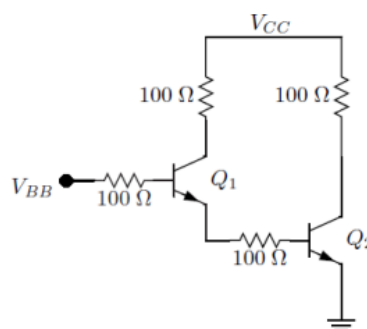
In the first experiment, We found β by setting up the experimental setup and dividing the current I_C by the current I_B .

$$V_{be} = 0,65 \text{ V} \quad V_{ce} = 1,98 \text{ V}$$

$$I_B = 0 \text{ A} \quad I_C = 5.1 \text{ A} \quad I_E = 5.1 \text{ A}$$



In the second experiment, The experimental setup was prepared and β_1 , β_2 , β_{TOT} were calculated.

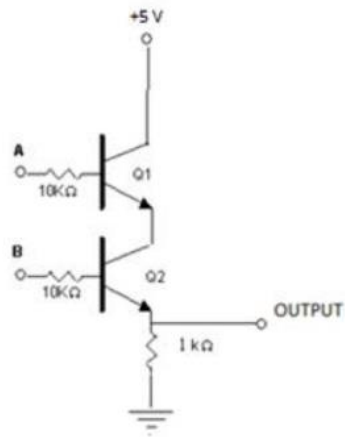


$$V_{BE1} = 643 \text{ mV} \quad V_{BE2} = 731 \text{ mV}$$

$$I_{B1} = 1 \quad I_{B2} = 1$$

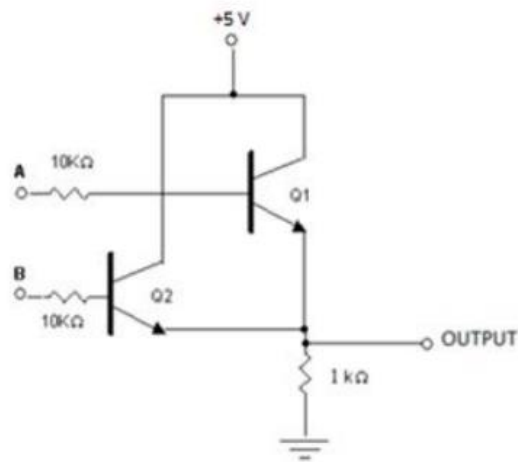
$$I_{C1} = 0.93 \text{ mA} \quad I_{C2} = 46.8 \text{ mA}$$

In the third experiment, in this experiment, we substituted the input values of A and B to get the results of the output.



V_A	V_B	V_{OUT}
0	0	0
0	5	0.43V
5	5	4.24V
5	0	0

In the fourth experiment, in this experiment, we substituted the input values of A and B to get the results of the output.



VA	VB	VOUT
0	0	0
0	5	4.30V
5	0	4.33V
5	5	4.33V

3. CONCLUSION & DISCUSSION

Bjt as a result of this experiment : Origin flowing from a terminal to a common terminal .It allows the current to increase the voltage and current by controlling the current flowing from the third terminal to the common terminal.

REFERENCES

- [1] IEEE Citation Style Guide, Available: <http://www.ijssst.info/info/IEEE-Citation-StyleGuide.pdf>, Access Date: October 2021